

HASIN AHMAD ZAFIR

Mechanical Engineering Student | Graduate Engineer Candidate

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PROFESSIONAL SUMMARY

Mechanical engineering graduate student at UNSW with hands-on experience across structural FEA, aerodynamic CFD, product design, and manufacturing. Founded and led Islamic University of Technology's first Formula SAE team, delivering a competition-ready car on an \$8,000 budget and placing 3rd at Formula Bharat 2023 against 53 teams. Experienced in wind-tunnel laboratory work, composite aerostructures, and engineering computation. Currently completing a thesis on geodesic winding path optimisation for composite pressure vessels. Seeking graduate and junior engineering roles in engineering consultancy, structural analysis, CFD/FEA, or Design.

CORE COMPETENCIES

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| <ul style="list-style-type: none">• Structural FEA — static, transient, composite laminates• Aerodynamic CFD — ANSYS Fluent, OpenFOAM• Filament winding & composite pressure vessel design• CAD/CAM design and DFM — SolidWorks, Fusion 360• Wind-tunnel scale model fabrication and instrumentation | <ul style="list-style-type: none">• Engineering computation — Python, MATLAB• Manufacturing: CNC, laser cutting, composite layup, lathe/milling• Technical documentation, thesis writing, client reporting• Team leadership and cross-functional project coordination• GD&T, tolerancing, BOM preparation, engineering drawings |
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FEATURED ENGINEERING PROJECTS

Formula IUT — FSAE Racing Car

Islamic University of Technology | Oct 2022 – Feb 2025

Founded and co-built IUT's first Formula SAE team, delivering a competition-ready spaceframe car on an \$8,000 budget. The team placed 3rd at Formula Bharat 2023 — Class 1 — against 53 competing teams.

Objective

- Design and fabricate a FSAE-compliant vehicle from inception to competition, meeting structural, safety, and performance regulations under resource constraints.

What I did

- Founded Formula IUT; recruited and managed cross-functional sub-teams across aerodynamics, suspension, drivetrain, and chassis
- Designed steel spaceframe chassis: factor of safety 1.8, final weight 98 kg
- Performed front-impact FEA and torsional stiffness analysis in ANSYS; validated against FSAE structural requirements
- Coordinated chassis fabrication using a welding jig; managed sponsor relations, \$8,000 budget, and competition logistics
- Presented engineering design documentation to judges at Formula Bharat 2023

Tools

SolidWorks · ANSYS FEA · MS Project · Steel fabrication

Outcome

3rd Place at Formula Bharat 2023 (Class 1) against 53 competing teams



Figure 1: Jig fixture for chassis

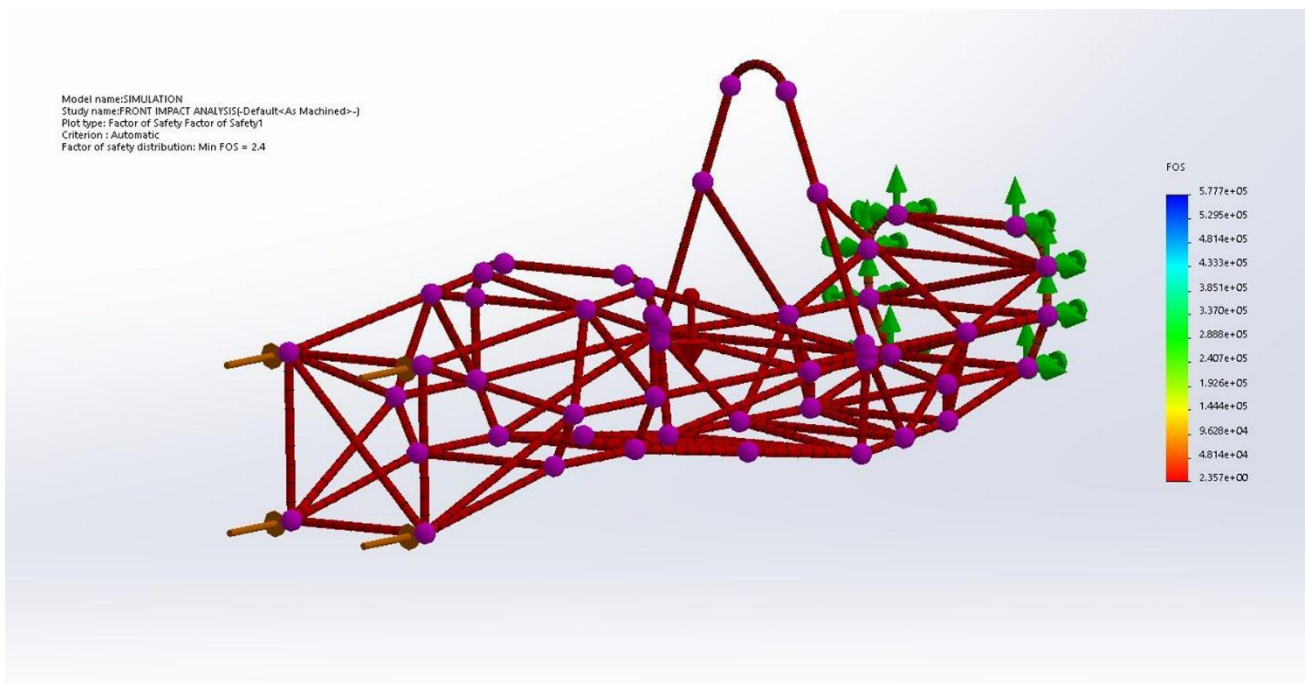


Figure 2: Front Impact Study



Figure 3: RS-01, first FSAE vehicle for Formula IUT

Design and Fabrication of an Impact Attenuator

Islamic University of Technology | Jan 2023 – Jan 2024

Designed a Formula Student-compliant frontal impact attenuator using recycled aluminium cans as the energy-absorbing medium. Validated via ANSYS transient FEA and UTM compression testing.

Objective

- Develop a low-cost, sustainable impact attenuator meeting FSAE energy absorption requirements (≥ 7.6 kJ) using recycled materials.

What I did

- Modelled attenuator geometry in SolidWorks using recycled aluminium cans as the crushing medium
- Ran transient crash FEA in ANSYS to predict crush behaviour, peak force, and energy absorption
- Validated design against FSAE specifications using UTM static compression testing
- Prepared technical report and engineering poster for competition review board

Tools

SolidWorks · ANSYS Transient FEA · UTM compression testing

Outcome

2nd Place out of 8 teams based on engineering design, simulation accuracy, and presentation; attenuator designed to absorb 7.6 kJ

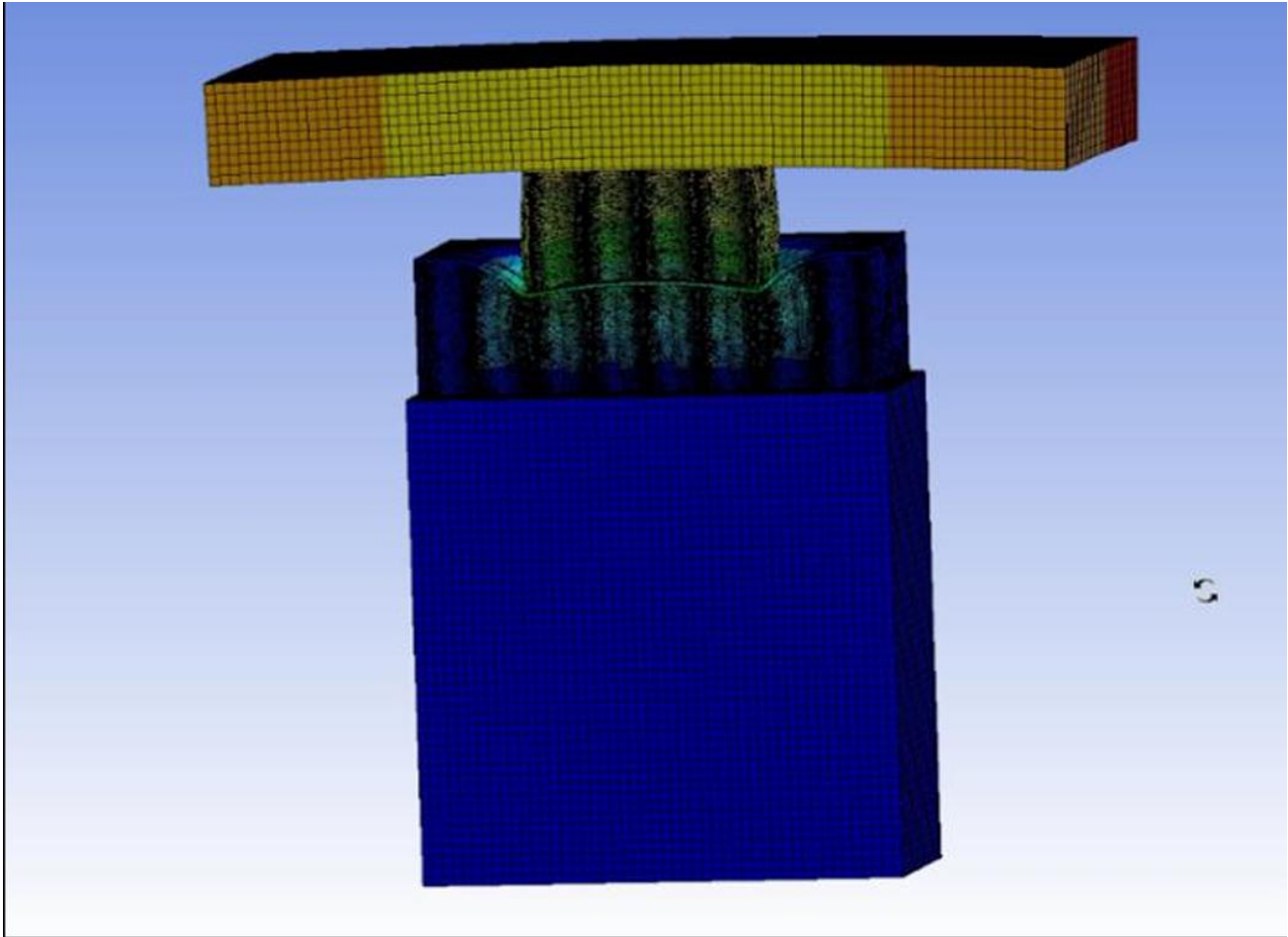


Figure 4: Transient Impact Analysis of Impact Attenuator

Design and Fabrication of a Stirling Engine

UNSW School of Mechanical & Manufacturing Engineering | Jun 2025 – Aug 2025

Client-based group project to design and fabricate a functional miniature Stirling engine for STEM outreach within 10 weeks and within budget.

Objective

- Deliver a working miniature Stirling engine prototype within an academic timeline and fixed budget, suitable for STEM demonstration to school-age audiences.

What I did

- Built a full 24-subassembly SolidWorks model; produced engineering drawings and bill of materials (BOM)
- Coordinated fabrication workflow across team members; applied tolerancing for dimensional fit and functional performance
- Managed project schedule using Gantt chart; prepared and delivered client-facing presentation
- Delivered prototype within 10 weeks at <1% dimensional deviation from the SolidWorks CAD assembly

Tools

SolidWorks · SolidWorks Drawing · Gantt planning · Lathe / milling

Outcome

Functional prototype delivered within 10 weeks; <1% dimensional deviation from CAD model; used for STEM outreach

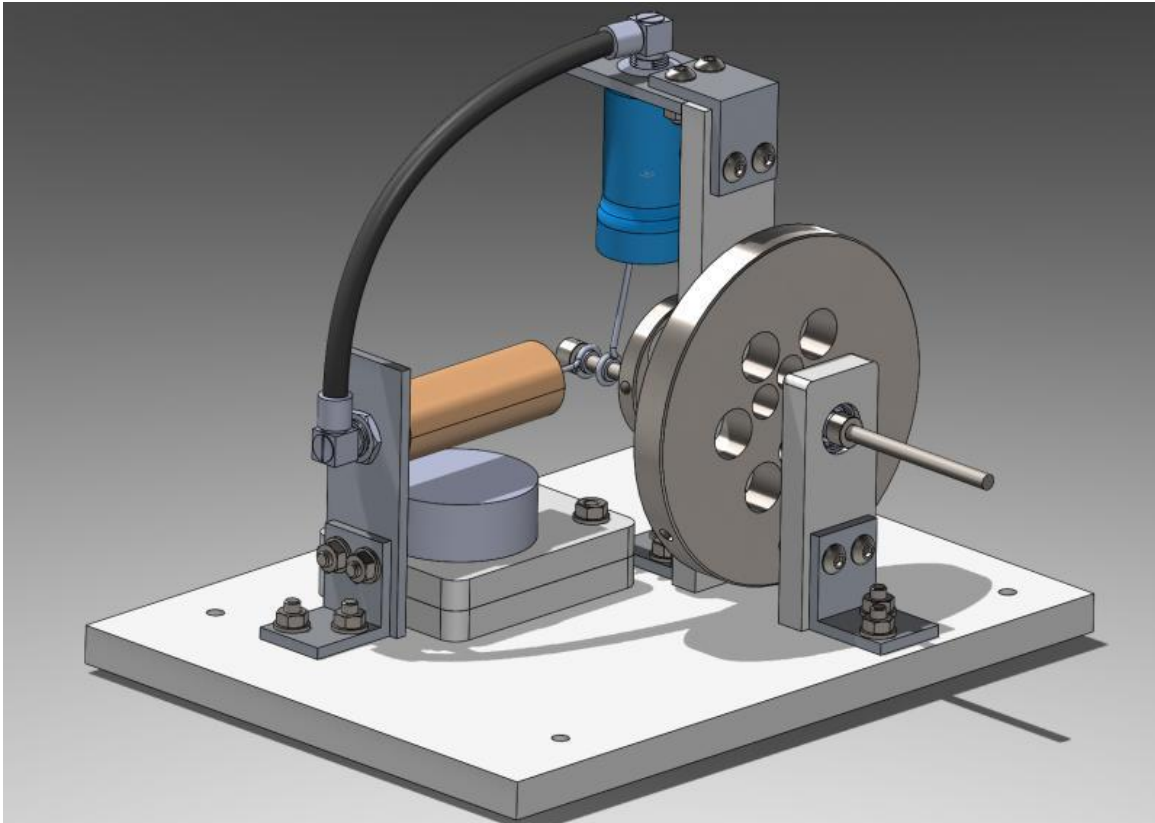


Figure 5: Designed Stirling Engine

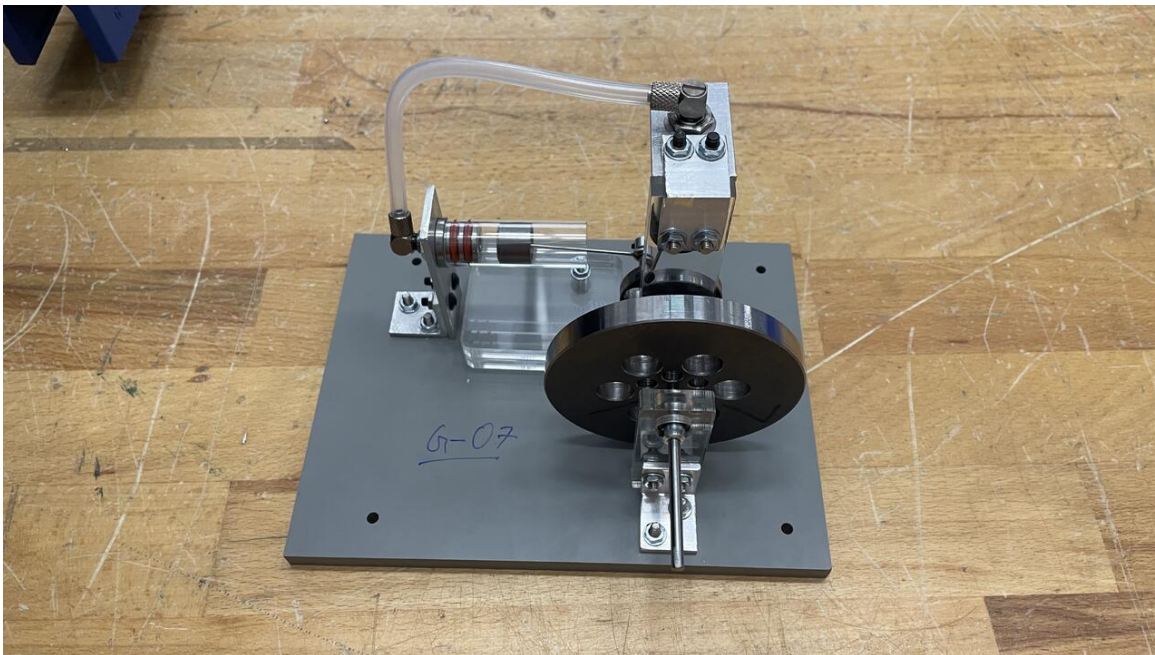


Figure 6: Assembled Stirling Engine

Miniature Swiss-Army Knife — CAD/CAM Manufacturing Project

UNSW School of Mechanical & Manufacturing Engineering | Sep 2025 – Nov 2025

Six-component miniature multitool designed and manufactured end-to-end using SolidWorks, Fusion 360, Denford 2D CNC, and injection-moulding trials.

Objective

- Demonstrate a complete CAD/CAM and DFM workflow: part design, CNC toolpath generation, mould fabrication, and physical production of a multi-component product.

What I did

- Modelled 1 of the 6 total components as part of a group; generated CNC toolpaths in Fusion 360 for Denford 2D CNC
- Applied DFM principles for draft angles, fit tolerances, and assembly across all components to ensure overall CNC operation time is kept under 2 minutes
- Fabricated moulds; conducted injection-moulding trials using Play-Doh to validate draft and tolerance performance
- Produced laser-cut acrylic prototypes as final fabricated output

Tools

SolidWorks · Fusion 360 · Denford 2D CNC · Laser cutting · DFM

Outcome

Six-component functional prototype manufactured; DFM and toolpath accuracy validated through full fabrication cycle



Figure 7: Mould after CNC operations

IMechE UAS Challenge — Aerodynamic Analysis & Fabrication

ANTS Aerial Systems | Feb 2021 – Jan 2022

Contributed to an award-winning fixed-wing UAS competition entry, performing CFD analysis of fuselage and payload configurations and supporting composite airframe fabrication.

Objective

- Design and fabricate a competition-compliant fixed-wing UAV optimised for aerodynamic performance and structural efficiency for the IMechE UAS Challenge.

What I did

- Performed ANSYS Fluent CFD analysis of fuselage and payload bay configurations to evaluate aerodynamic drag and flow separation
- Contributed to airframe fabrication: laser cutting, foam shaping, and composite layup
- Supported final assembly and pre-competition integration testing

Tools

ANSYS Fluent · SolidWorks · Composite layup · Ultimaker Cura · Laser cutting

Outcome

IMechE UAS Challenge Design Award 2021

Iteration = 188
33.617
31.123
28.729
26.335
23.941
21.546
19.152
16.758
14.364
11.970
9.576
7.182
4.788
2.394
0
Velocity [m/s]
Cut Plot 1: contours

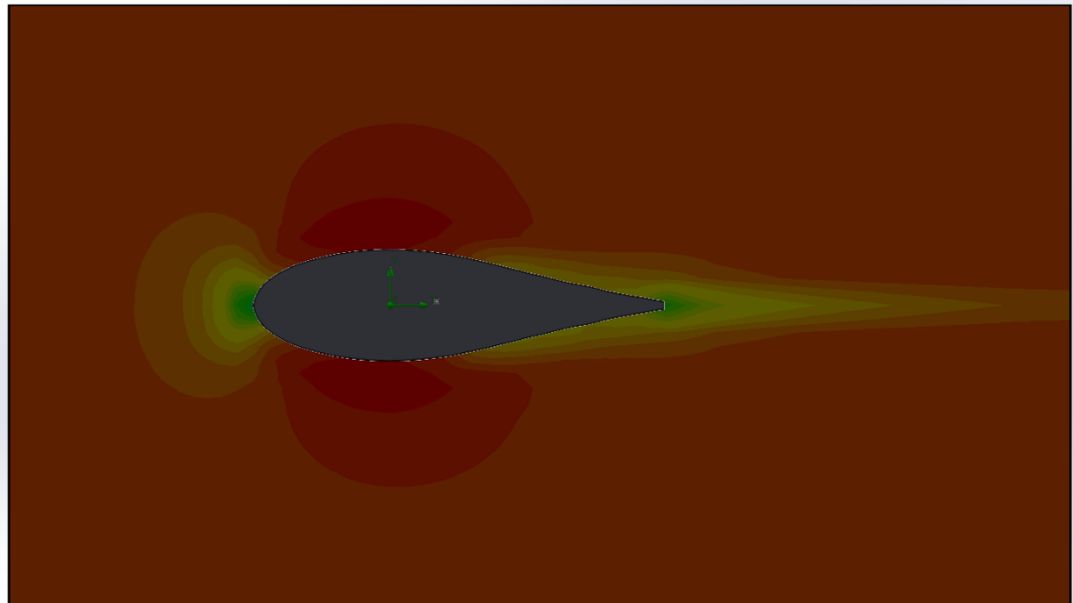


Figure 8: Aerodynamic Simulation for payload carrier